

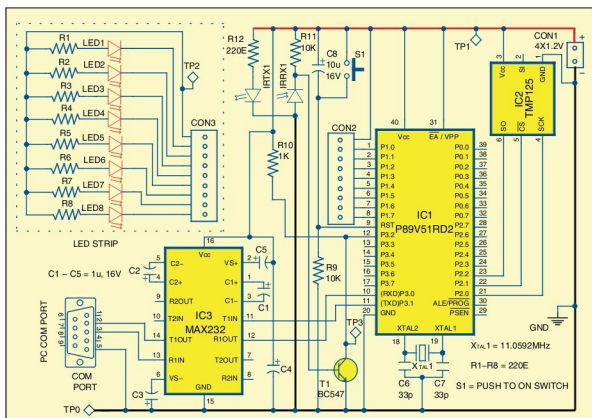


Fig. 3: Circuit of the propeller message display with temperature indicator

microcontroller P89V51RD2 (IC1), temperature sensor TMP125 (IC2), MAX232 (IC3) and a few discrete components. Port pins P1.0 through P1.7 are connected to CON2, which further needs to be connected to CON3 of the LED strip comprising LED1 through LED8. These LEDs form a circular display when rotated very fast.

Microcontroller. The heart of the system is microcontroller P89V51RD2. The P89V51RD2 is an 8-bit 80C51 microcontroller with 64KB Flash and 1024 bytes of data RAM. It features three 16-bit timers/counters, programmable watchdog timer, eight interrupt sources with four priority levels, serial peripheral interface (SPI) and enhanced UART, programmable counter array with PWM and capture/compare functions, and four 8-bit input/output (I/O) ports. It has an on-chip oscillator and clock circuitry, which is operated up to 40MHz crystal. It supports 12 clocks per machine cycle (default) and 6 clocks per machine cycle mode, which can be selected via software. The Flash program memory supports both parallel programming as well as serial programming.

Power-on reset is provided by the combination of resistor R9 and capacitor C8. Switch S1 is used for manual reset. An 11.0592MHz crystal (X_{TAL1}) along with two 33pF capacitors (C6 and C7) provides basic clock frequency to the microcontroller.

Temperature sensor with digital output. TMP125 is an SPI-compatible temperature sensor available in tiny SOT23-6 package. Requiring no external components, it is capable of measuring temperatures within accuracy of 2°C over a temperature range of -25°C to +85°C and within 2.5°C accuracy over -40°C to +125°C. Low supply current and a supply range from 2.7 V to 5.5 V make TMP125 an excellent candidate for low-power applications. TMP125 is ideal for extended thermal measurement in a variety of communication, computer, consumer, environmental, industrial and instrumentation applications. Port pins P2.0 through P2.2 of IC1 are used to interface the temperature sensor as shown in Fig. 3.

Interrupt sensor. A sensor is built (see Fig. 4) using IR transmitter (IRTX1) and receiver (IRRX1) to

generate interrupts. In each revolution, as the beam is interrupted, the sensor generates a positive pulse, which is inverted with the help of a BC547 npn transistor (T1). The inverted pulse is fed to controller port pin P3.2. The microcontroller executes an interrupt routine when a pulse arrives.

The interrupt routine measures the total time taken for one revolution, divides it by 360 and stores the result in another timer, configured in auto-reload mode. Now the timer overflows after tiny intervals of time. Each time it overflows,

the next stored value is called from the lookup table and displayed (refer Fig. 5).

Alternatively, sensor MOC7811 can be used in place of the arrangement

Test Points

Test point	Details
TP0	GND, 0V
TP1	4.8V (when batteries used)
TP2	4.8V (when batteries used)
TP3	High to low when beam interrupted

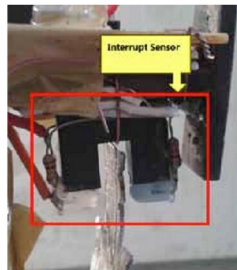


Fig. 4: Interrupt sensor